

METALS AND NON – METALS

▪ Elements

- Elements are purest form of matter which can neither be created by two or more substance nor be split into two or more substance by any physical and chemical process.

Eg:- Hydrogen, Helium, Lithium, Beryllium, Potassium, Sodium, Copper, Iron, etc.

There are 118 elements are discovered at present time.

▪ Properties of elements

- Element are made by similar type of atoms.
- The smallest particle of an element is called atom.

▪ Types of element

1. Metal
2. Non – metal
3. Metalloids

1. Metals

- Those elements which have 1 to 3 electrons in their outermost shell called metal.
- The metal generally contain “um” suffix.

Eg:- Lithium (Li), Beryllium (Be), Potassium (K) / Kalium, Sodium (Na) / Natrium, Copper (Cu) / Cuprum, Iron (Fe) / Ferrum, Argentum (Ag) / Silver, Gold (Au) Aurum, Calcium (Ca), Gallium (Ga), Chromium (Cr), Zinc (Zn), Cobalt (CO).

▪ Properties of metals

Case 1: Physical properties of Metals

- Metals are shiny which is known as metallic lustre. So, Metals are lustrous.
- Metals are hard & rigid
Exception:
 - ✓ Sodium & Potassium are soft which can be cut by knife.

- Metals are Solid at room temperature.
Exception:
 - ✓ Mercury (Hg), Gallium [M.P = 29.8°C]

- Metals are ductile i.e, they can be made into thin wires by stretching.
Exception:
 - ✓ Mercury, Sodium, Potassium & Gallium etc.

Example:-

- ✓ 2 Km long wire can be formed by 1g gold.

- Metals are malleable i.e, they can be beaten into thin sheets.
Exception:
 - ✓ Mercury, Na, K, Ga

Example:-

- ✓ Gold, Silver, etc.

- Metals have high melting point & boiling point.

Exception:

- ✓ Mercury & Gallium [29.8°]

- The density of metal is high because mass (↑) & volume (↓).

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Exception:

- ✓ Li, Na, K have low densities.

- Metals are good conductor of electricity and heat because they have free electrons.

Example:-

- ✓ Silver is best conductor of electricity.
✓ Lead is least conductor of electricity.

- Metals are sonorous i.e, makes ringing sound on biting.

- Metals are electropositive.

- Metals are electron's donor.

Case 2: Chemical properties of metals

- Reaction of metal with oxygen

- Some metals react with oxygen to form metal oxide which is basic in nature.



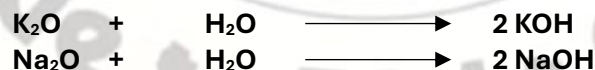
Example:-



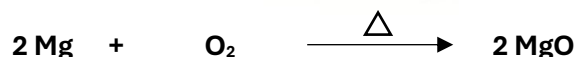
Note:

1. Potassium and Sodium are stored under kerosene oil to prevent their reaction with the oxygen moisture and carbon – dioxide of air because sodium burns in oxygen to form sodium oxide.

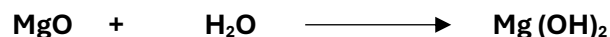
2. Most of the metals oxide are insoluble in water but some of the metal oxide dissolve in water to form alkali.



3. Magnesium metals do not react with oxygen at room temperature but on heating it reacts with oxygen to form magnesium oxide.



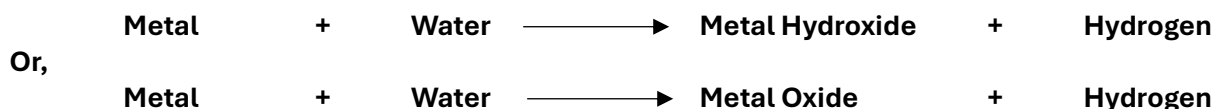
Magnesium oxide partially dissolve in water to form magnesium Hydroxide Solutions.



4. Aluminium metal reacts with oxygen on heating to form Aluminium oxide.



▪ Reaction of metal with water to form metal hydroxide or metal oxide and hydrogen.



Example:-

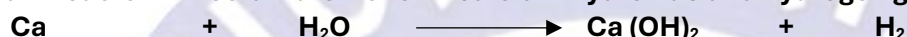
- ✓ When potassium reacts with cold water to form potassium hydroxide and hydrogen gas.



- ✓ Sodium reacts with cold water to form sodium hydroxide and hydrogen gas.



- ✓ Calcium reacts with cold water to form calcium hydroxide and hydrogen gas.



Note:

1. Magnesium do not react with cold water it react with hot water [Steam].



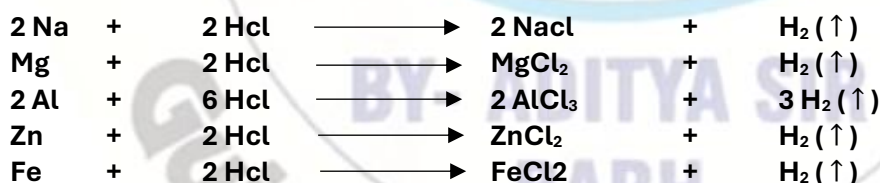
2. Aluminium & Iron react with steam only.



▪ Reaction of metal with dilute acid



Eg:-



Note:

1. Copper, Silver & Gold do not react with dilute acid.
2. Evolution of hydrogen gas in the reaction of metal and dilute acid due to metals are more reactive than hydrogen that means loose electron more easily than hydrogen in fact the more reactive metals give electrons easily and this electron reduce the hydrogen ions of acid to hydrogen gas.

• Aqua Regia

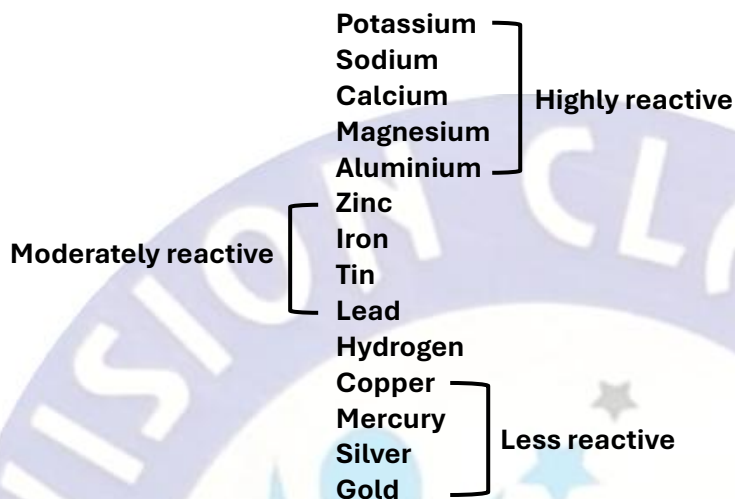
- Aqua regia is a homogeneous mixture of one part of Nitric acid and three part of concentrated hydrochloric acid.

i.e, $\text{HNO}_3 : 3\text{HCl}$

1 : 3

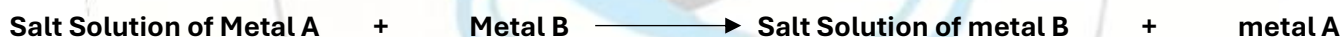
Formula: $\text{HNO}_3 + 3\text{HCl}$

- **Reactivity Series**

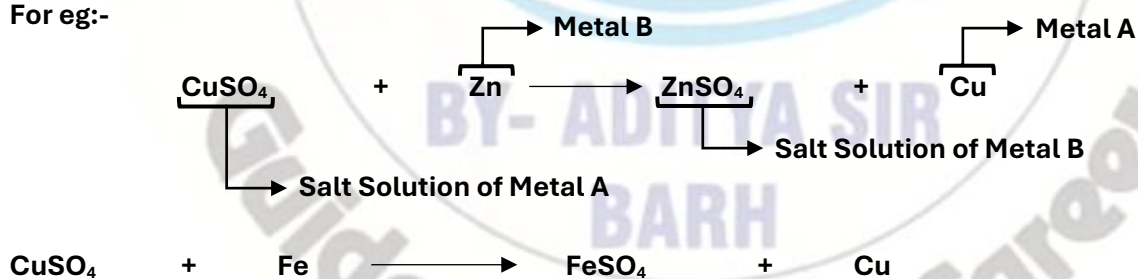


- **The strength of reactive metals depend on the donating power of electrons.**

- **When a more reactive metal displace less reactive metal from its salt solution.**

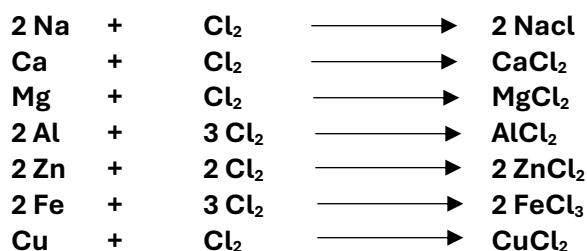


For eg:-



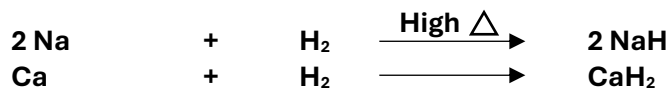
- **Metal reacts with chlorine to form metal chloride.**

For eg:-



▪ **Reaction of metal with hydrogen**

- Metal do not react with hydrogen only some metals like sodium, Potassium, magnesium reacts with hydrogen to form metal hydride.



Non – Metals

- Those elements which have 5 to 8 electrons in their outermost shell are called Non – metals.

For example:-

- ✓ Nitrogen (N₂), Oxygen (O₂), Fluorine (F₂), Phosphorus (P₄), Sulphur (S₈), Chlorine (Cl₂), Bromine (Br₂), Iodine (I₂) Selenium (Se).

Exception:

- ✓ Carbon has 4 electrons in outermost shell.

- All inert gases are non – metal.

For example:-

- ✓ Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe), Redon (Rn).

▪ **Properties of Non – metals**

Case 1: Physical properties of non – metals

1. Non – metals are not lustrous.

Exception:

- ✓ Graphite & Iodine.

2. Non – metals are brittle.
3. Non – metals are not ductile.
4. Non – metals are not sonorous.
5. Non – metals are poor conductor or insulator of electricity.

Exception:

- ✓ Graphite is conductor of electricity.

6. Non – metals are solid, liquid & gas at room temperature.
7. Non – metals have low melting and boiling points.
8. Non – metals have less density than metal.
9. Non – metals have less hardness than metal.

Exception:

- ✓ Diamond is the hardest substance in the universe.

10. Non – metals have low tensile strength.

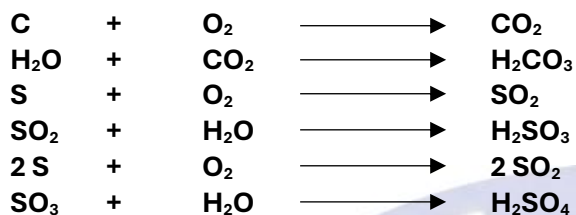
→ They can not hold heavy weight

11. Non – metals are electronegative.
12. Non – metals are electron acceptor.

Case 2: Chemical Properties of Non – metals

- Reaction of non – metal with oxygen to form acidic oxides or neutral oxides.

Eg:-



- Carbon monoxide (CO) & Nitrous oxide (N₂O) are neutral oxides

- Reaction of non – metal with water

- Non – metals do not react with water.

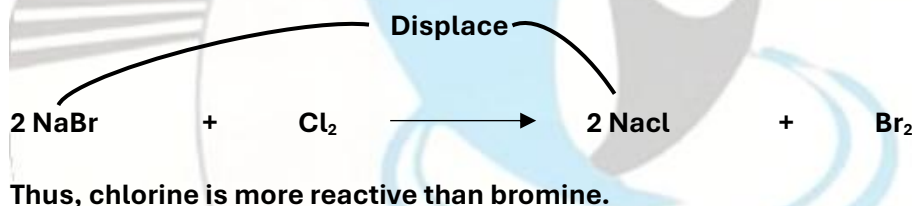
- Reaction of non – metal with dilute acid

- Non – metals do not react with dilute acid.

- Reaction of Non – metal with salt solution

- A more reactive non – metal displace a less reactive non – metal from its salt solution.

For Eg:-

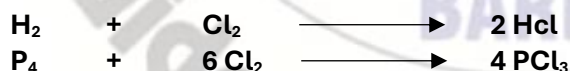


- Reaction of Non – metal with chlorine

- Non – metal reacts with chlorine to form covalent chlorides which are non – electrolytes. [don't conduct electricity].

- Non – metals chlorides are usually liquid or gases.

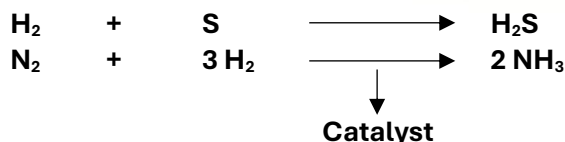
For eg:-



- Reaction of non – metal with hydrogen

- Non – metals react with hydrogen to form covalent hydrides.

For eg:-



- **Catalyst**

- That substance which either increase or decrease the rate of chemical reaction.

+ve catalyst: Increase rate of reaction

-ve catalyst: Decrease rate of reaction

▪ **Reactive series of non – metal**

Fluorine	↓ Decreases
Chlorine	
Bromine	
Iodine	
Astatine	

• **Electronic configuration**

- An arrangement of electrons of an element in shell and sub – shell is called electronic configuration.

Shell	No. of maximum electrons = $2n^2$
n = 1, K	$2 \times 1^2 = 2$
n = 2, L	$2 \times 2^2 = 8$
n = 3, M	$2 \times 3^2 = 18$
n = 4, N	$2 \times 4^2 = 32$
n = 5, O	$2 \times 5^2 = 50$

Where, n is position of orbit.

• **Valence electrons (s)**

- The outermost shell electrons present in an element is called valence electrons.

▪ **Elements and their valence e^- (s)**

Elements	Elements Name	Valence Electrons
1	H	1
2	He	2
3	Li	1
4	Be	2
5	B	3
6	C	4
7	N	5
8	O	6
9	F	7
10	Ne	8
11	Na	1
12	Mg	2
13	Al	3
14	Si	4
15	P	5
16	S	6
17	Cl	7
18	Ar	8
19	K	1
20	Ca	2

• **Ions**

- The positive and Negative charge contain atom called Ion.

OR

It is an electrically charge atom or molecule formed by loss or gain of electron.

For eg:-

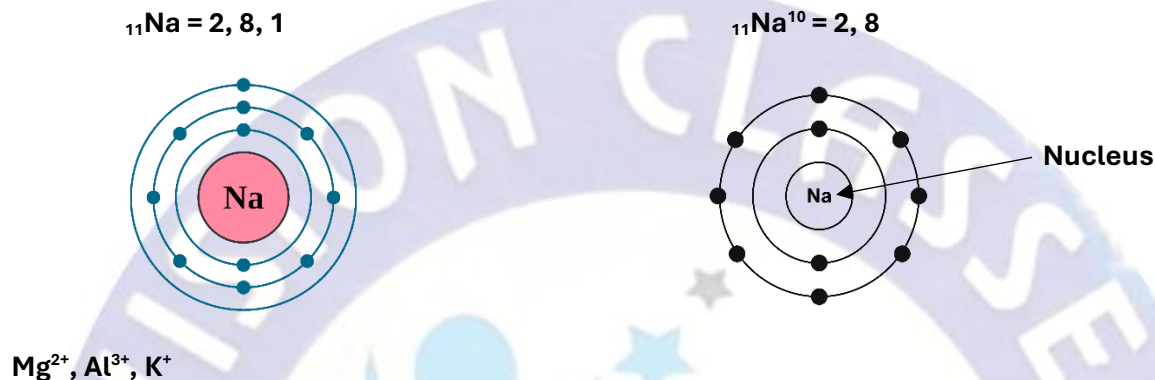
✓ Ca^{2+} , Mg^{2+} , Al^{3+} , N^{3-}

Types of Ions

1. Cation

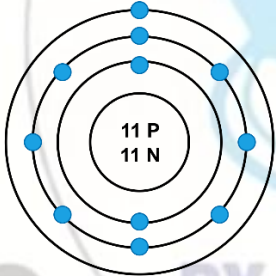
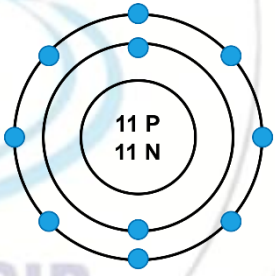
- Metal forms cations by losing its valence electron.
- Cations are positively charged particles.

For eg:-



▪ Difference between metal and its cation

Eg - Na

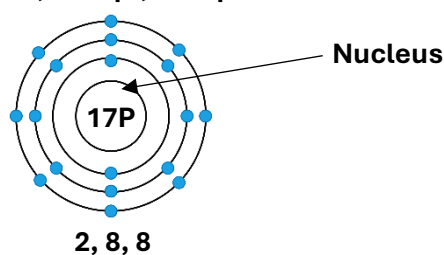
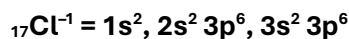
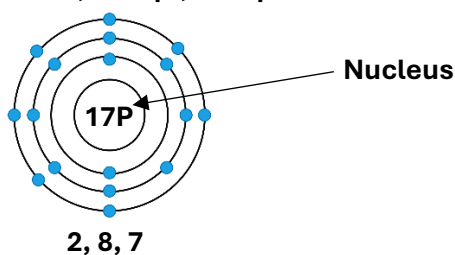
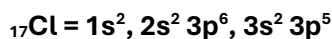
Metal	Its Cation
${}_{11}\text{Na} = 1s^2, 2s^2 2p^6, 3s^1$	${}_{11}\text{Na}^{+} = 1s^2, 2s^2 2p^6$
	
It has more no. of electrons.	It has less no. of electrons.
Its size is large.	Its size is small.
It is electrically neutral.	It is positively charged.
It is less stable.	It is highly stable.
It has more no. of orbits.	It has less no. of orbits.

2. Anion

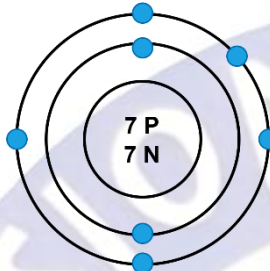
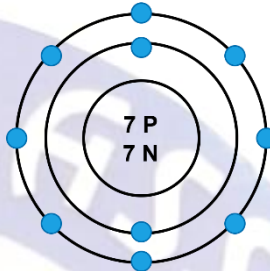
- It is formed by non-metal by gaining electrons.

For eg:-

✓ N^{3-} , O^{2-} , F^{-} , S^{2-} , P^{3-} , Cl^{-} etc.



▪ **Difference between non – metal & its Anion**

Non – metal	Its Anion
It have less no. of electrons.	It have more no. of electrons.
Its size is small.	Its size is large.
It is less stable.	It is more stable.
It is electrically neutral.	It is negatively charged.
It have 7 electrons in their outermost shell.	It have 8 electrons in their outermost shell.
${}_7\text{N} = 1s^2, 2s^2 2p^3$  Nitrogen [2, 5]	${}_7\text{N}^{3-} = 1s^2, 2s^2 2p^6$  Nitrogen Ion [2, 8]

• **Lewis Dot Structure OR Electron dot Structure**

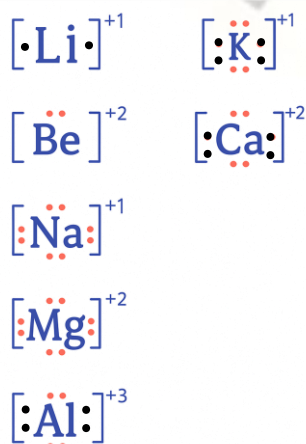
- A representation of the valence electrons of an atoms that uses dots around the symbol of the element is known as Lewis dot structure.

Case 1: Lewis dot structure of atom

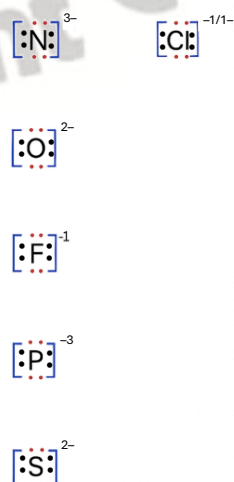
H•	Na•
He:	:Mg
Li•	:Al•
Be:	:Si
:B•	:P:
:C•	:S:
:N:	:Cl•
:O:	:Ar:
:F•	K•
:Ne:	:Ca

Case 2: Lewis dot structure of ions

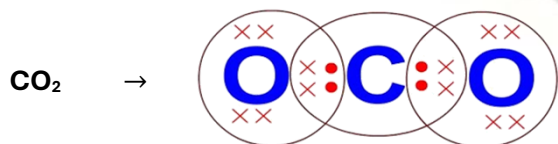
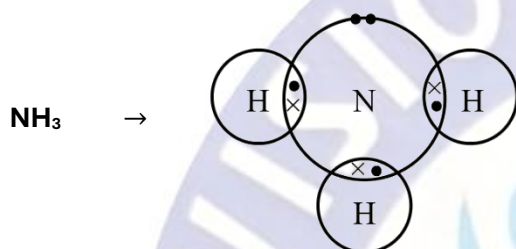
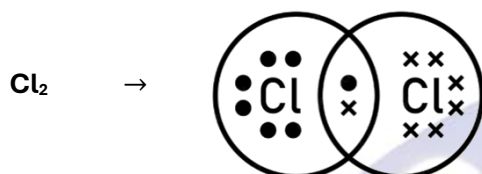
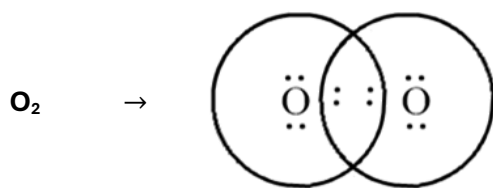
Cations



Anions



Case 3: Lewis dot structure of molecule



Chemical Bonding

• **Bond / Chemical Bond**

➤ That attraction force which binds atoms together in a molecule is called Chemical Bond.

▪ **Mechanism of chemical bonding**

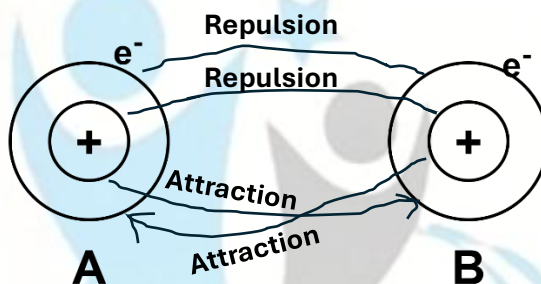
1. How do atoms combine ?
2. Why do atoms combine ?
3. What happens when they combine ?

▪ **Conclusion for (i)**

Let A & B two isolated atoms



1. When the two atoms A & B are separated by a large distance then there is no interaction between A and B.
2. When the atoms approach to each other then four types of forces come into play.



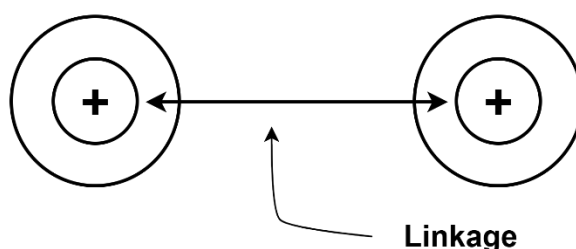
3. If the magnitude of attraction force is greater than repulsion then the formation of the molecule takes place.
4. If the magnitude of repulsion force is greater than attraction force then formation of the molecule does not take place.
5. Atom loses its internal energy because of the atom towards the force of attraction.
6. Formation of molecules is exothermic process.
7. Formation of molecule is a chemical process.
8. Stability of a molecule to energy release.
9. Potential energy of the molecule is less than its constituents.

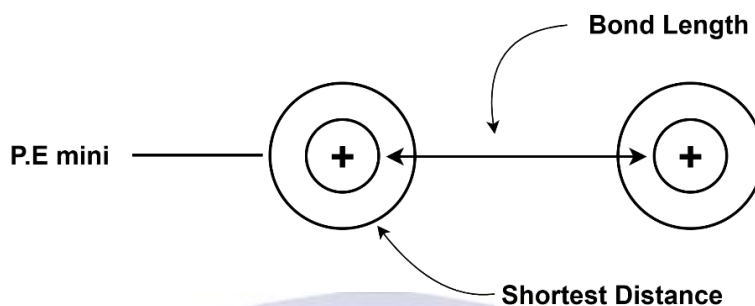
• **Bond length**

- It is defined as, the shortest distance between the nuclei of two atoms where the value of potential energy is minimum.

• **Bond**

- It is a linkage which is satisfied between the two atoms when they combine chemically.





▪ **Process of formation of chemical bonding for high stability**

1. Transfer of valence electrons.
2. Sharing of valence e^- (s).
3. Donation as well as sharing of valence e^- (s).

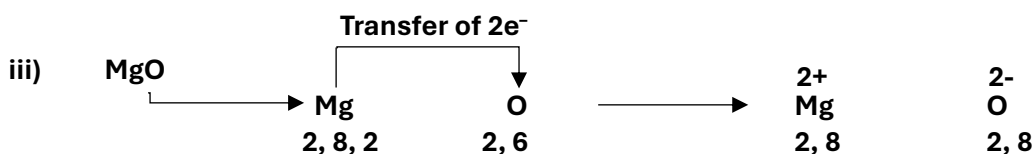
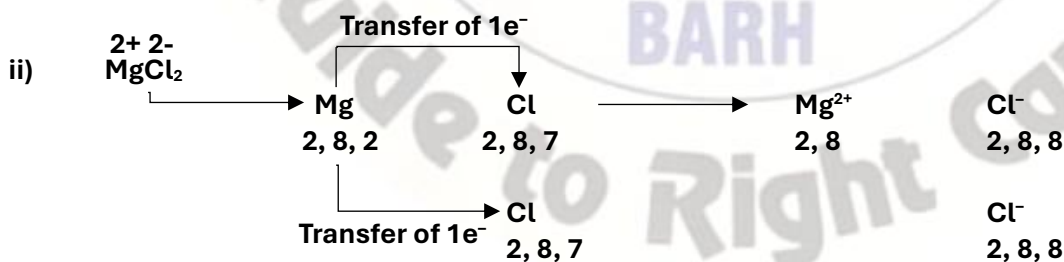
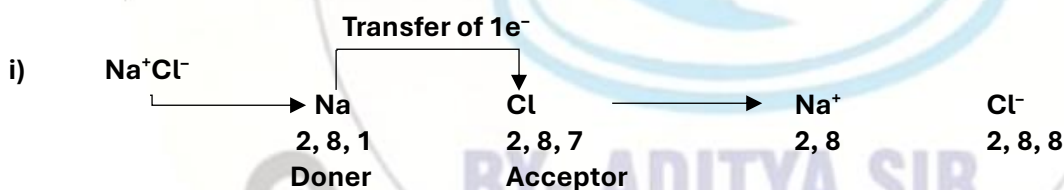
Types of Chemical bond

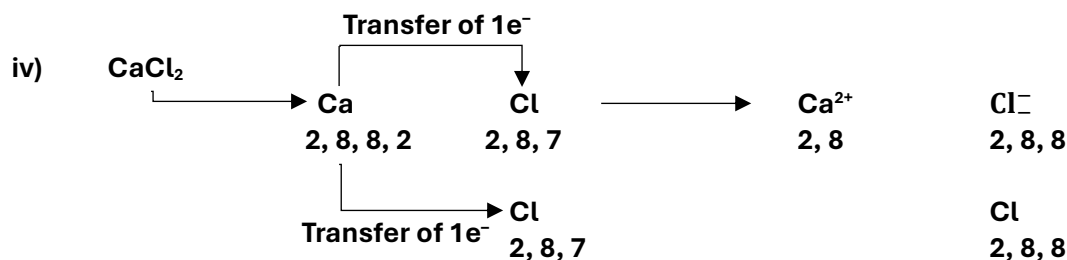
1. Ionic / Electrovalent / Polar Bond.
2. Covalent Bond.
3. Coordinate Bond or Semi – Polar Bond.
4. Singlet Bond.

1. Ionic / Electrovalent Bond

- This bond is formed by completely transfer of valence electron.
- This bond is formed between metal and non – metal.
- This bond is non – directional.

For eg:-



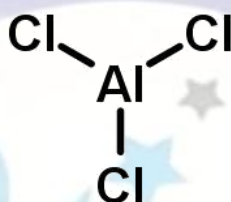


Other eg of ionic bond:-

✓ KNO_3 , CuSO_4 , NH_4Cl .

Note:

➤ Aluminium chloride [AlCl_3] do not form ionic bond. It forms covalent bond.



▪ Properties of ionic compounds

1. These are form between cation & Anion.
2. They have high melting and boiling point because they are bonded by strong ionic bond.
3. Ionic compound are hard, brittle, solid.
4. The aqueous solution of ionic compound conduct electricity.
5. Ionic compound do not exhibit in phenomenon of isomerism.

• Covalent Bond

- The bond which is formed by sharing of valence electron between atoms are called covalent bond.
- This bond formed between non – metal and non – metal.
- It is directional i.e, an angle is fixed between atoms and molecules.
- This bond is weaker than ionic bond.

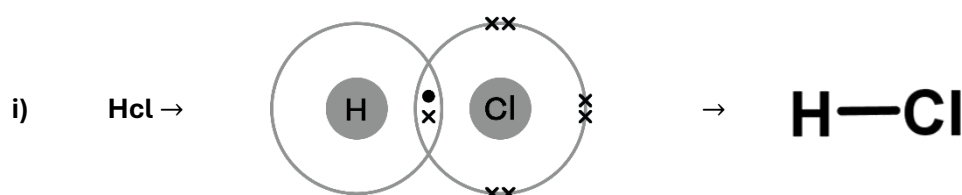
Types of Covalent Bond

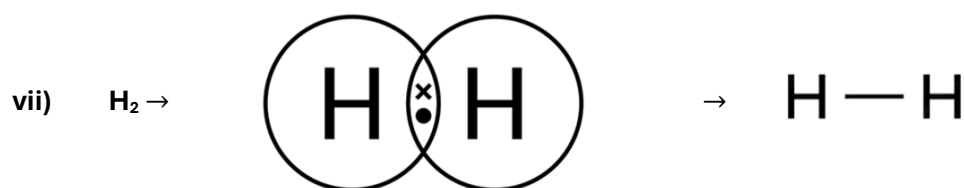
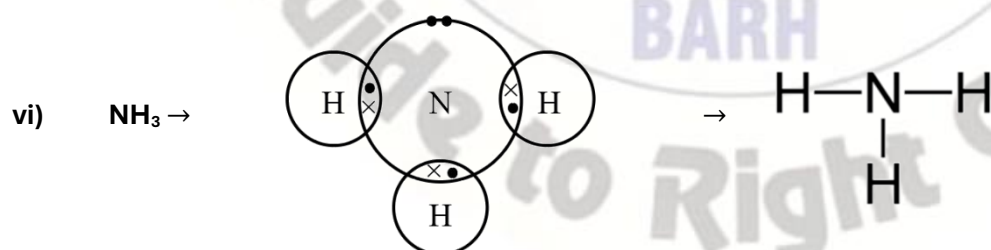
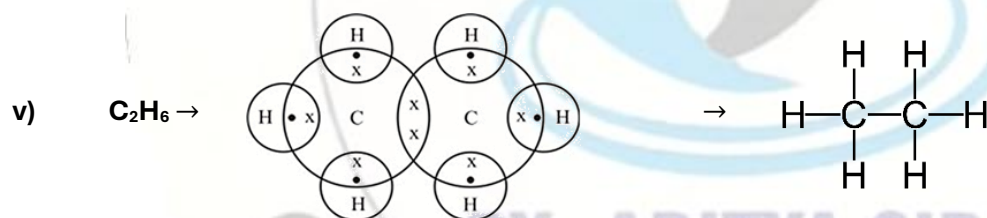
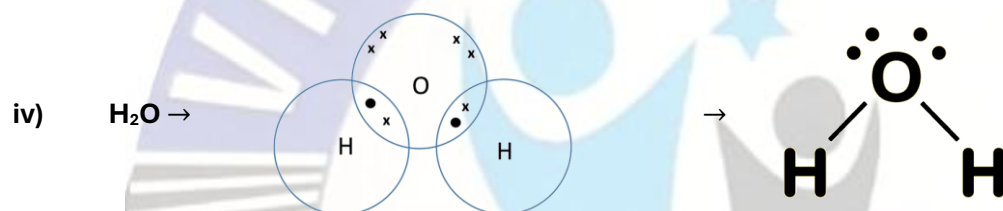
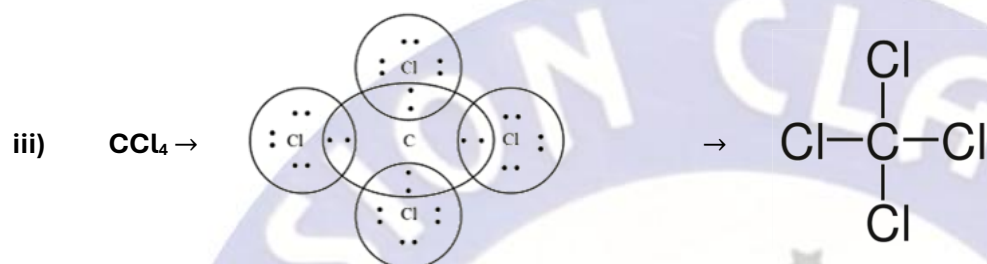
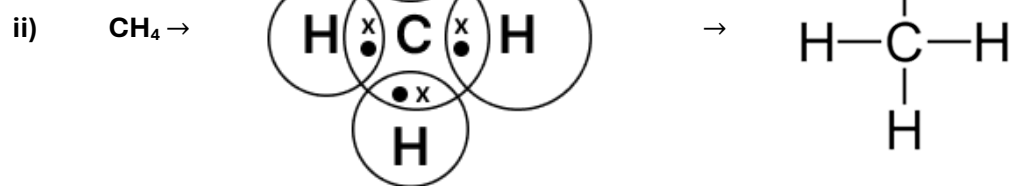
1. Single Covalent bond.
2. Double Covalent bond.
3. Triple Covalent bond.

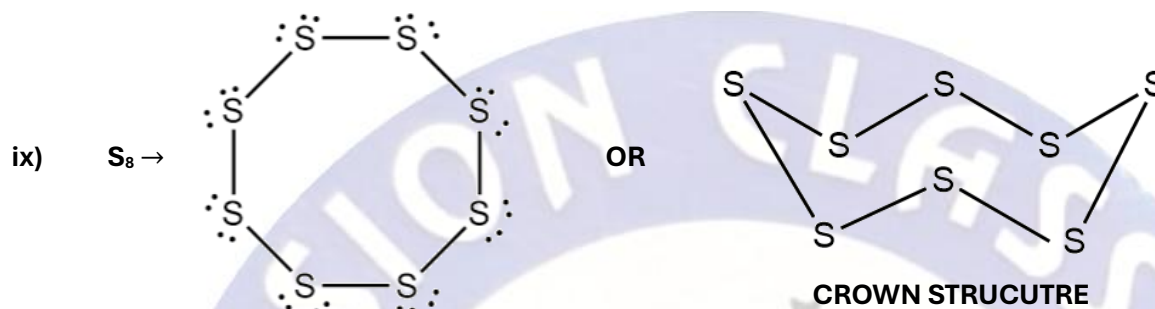
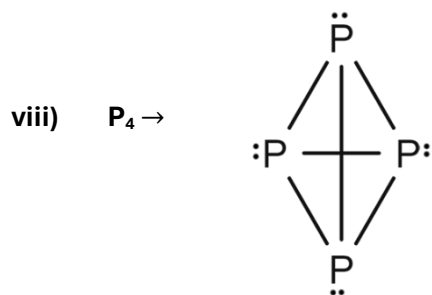
1. Single Covalent bond

- The bond which is form by sharing of a pair of electrons, is called Single Covalent bond.

For eg:-



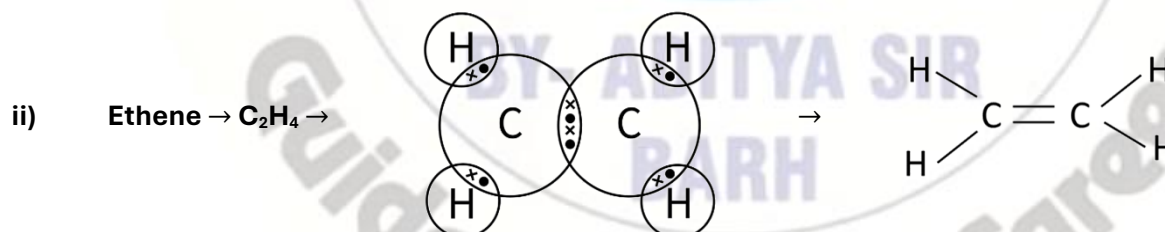
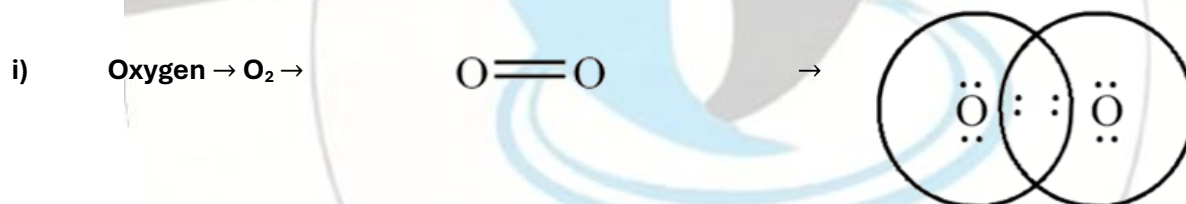




2. Double Covalent bond

➤ That bond is formed by sharing of two pair of lone electrons, is called Double Covalent bond.

For eg:-



3. Triple Covalent bond

➤ That bond which is formed by sharing of three lone pair of electron, called triple covalent bond.

For eg:-

✓ Ethyne / Acetylene $[C_2H_2] \rightarrow H - C \equiv C - H$

✓ Nitrogen $[N_2] \rightarrow N \equiv N$

Properties of Covalent bond

1. They are soft, easily feasible in solid, liquid & gas.

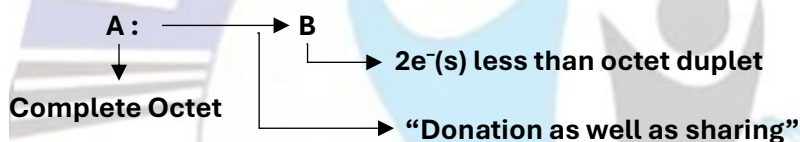
2. They have low melting & boiling point than ionic bond.
3. They are springly soluble in water but, fearly soluble in non – polar solvent.
4. Covalent bond is directional so covalent compound shows phenomenon of isomerism.
5. Their aqueous solution is poor conductor of electricity because they do not ionised in water.

▪ Difference between ionic compound and covalent compound

Covalent Compound	Ionic Compound
They are springly soluble in organic solvent.	They are not soluble in organic solvent.
They do not ionize.	They are ionize in its solution or molten solution.
They are bad conductor of heat and electricity.	They are good conductor of heat and electricity.
They have weak force of attraction between molecule.	They have strong force of attraction between molecule.
Formed by sharing of electrons.	Formed by transfer of electrons.
Generally have low melting & boiling point.	It has high melting & boiling point.

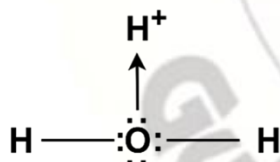
Coordinate Bond

• Introduction

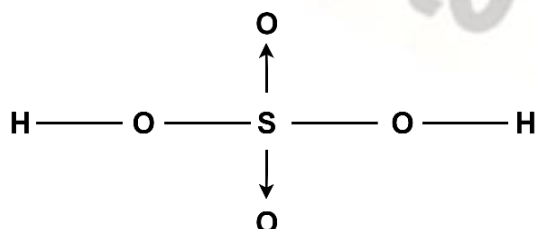


For example:-

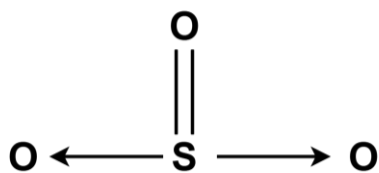
i) Hydronium ion (H_3O^+)



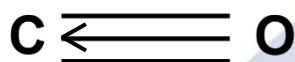
ii) Sulphuric acid (H_2SO_4)



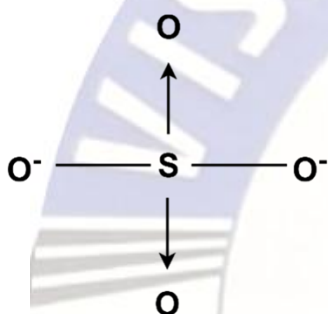
iii) Sulphite (SO_3)



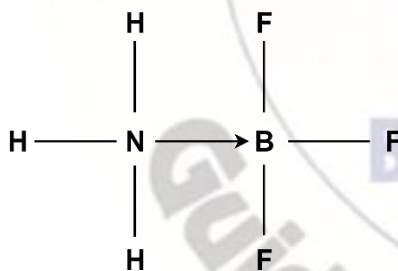
iv) Carbon monoxide (CO)



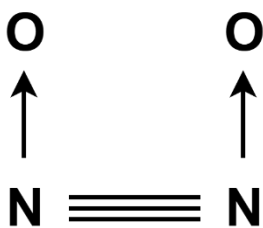
v) SO_4^{2-}



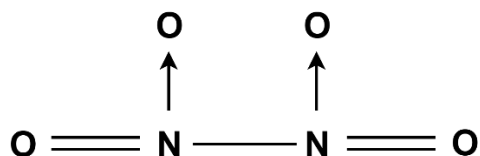
vi) NH_3BF_3



vii) N_2O_2



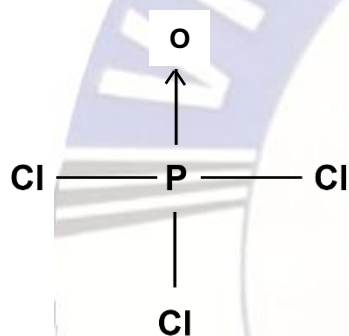
viii) N_2O_4



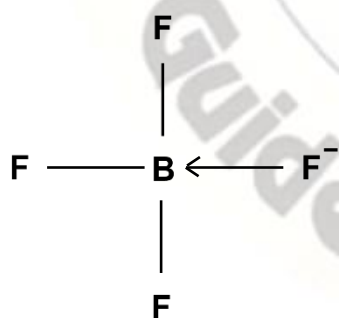
ix) HClO_4



x) POCl_3



xi) BF_4^-



▪ How do select the Central atom of a molecule

⇒ There are three conditions for selecting central atoms.

1. Less number.
2. Higher covalency [No. of shared (e^-)s $\uparrow$]
3. Size of atom.

- Failure of Octet Rule**

- Many examples are known in which same atoms bonded to other atoms by a covalent bond have either less than 8 electrons [incomplete octet] or more than 8 electrons [expansion of octet].

- Incomplete Octet [Less than $8e^-(s)$]**

Compound	Structure	Surrounding $e^-(s)$ of central atom
BCl_3	<pre> Cl — B — Cl Cl </pre>	6
$AlCl_3$	<pre> Cl — Al — Cl Cl </pre>	6

- Expansion of Octet [more than $8e^-(s)$]**

Compound	Structure	Surrounding $e^-(s)$ of central atom
PCl_5	<pre> Cl Cl — P — Cl \ Cl Cl </pre>	10
IF_7	<pre> F F — I — F / \ / \ F F F F F </pre>	14
SF_6	<pre> F F — B — F / \ / \ F F F F F </pre>	12

- According to the above result there are two theory have been developed.

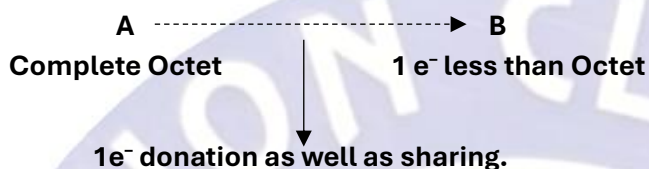
1. Concept of Single Bond [Given by Sudgen].
 [In the favour of Octet]

2. Sidwick rule of maximum covalency.
 [Non – supporter of Octet]

- **Singlet Bond**

- Given by Sudgen.

- This bond is formed between two atoms by sharing of one electron.

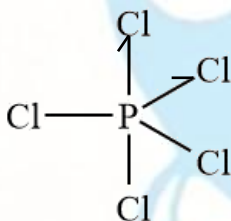


- It is also called “Single electron linkage” or “half bond”.

————— OR ————— OR —————>

For example:-

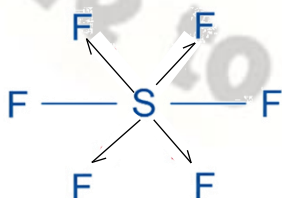
✓ PCl_5



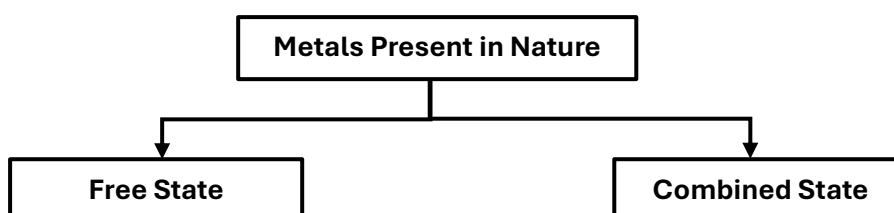
✓ IF_7



✓ SF_6



Metallurgy



1. Free State

- Those metal which do not react with moisture, air, water, etc. are called free state metal.

For example:-

- ✓ Platinum, Gold, Silver, etc.

2. Combined State

- Those metal which react with moisture, air, water, etc are called combined state metal.

For example:-

- ✓ Aluminium, Magnesium, Iron, Calcium, Sodium.

• Minerals

- Those natural substance which found in earth crust in which the metals and their compound are found are called minerals.

• Ores

- Those minerals from which the metals can be extracted correctly and profitably are called ores.

Note:

- Those minerals which have in rich amount of metal are called ores.
➤ All the ores are called minerals, but all the minerals are not called ores.

<u>Metal to be Extracted</u>	<u>Name of Ores</u>	<u>Name of Compound</u>	<u>Formula of Ores</u>
Sodium	Rock Salt	Sodium Chloride	NaCl
Aluminium	Bauxite	Aluminium Oxide	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
Manganese	Pyrolusite	Manganese Oxide	MnO_2
Zinc	Calamine	Zinc Carbonate	ZnCO_3
	Zinc Blende	Zinc Sulphide	ZnS
Iron	Hematite	Iron (III) Oxide	Fe_2O_3
Copper	Cuprite	Copper (I) Oxide	Cu_2O
Mercury	Cinnabar	Mercury (II) Sulphide	HgS

Types of Ores

a) Oxide Ore

- Copper, Aluminium, Zinc, Tin, Iron, etc occurs as Oxide.

b) Sulphide Ore

- Silver, Copper, Zinc, Mercury, lead, Iron, etc occur as Sulphide.

c) Carbonate Ore

- Sodium, Copper, Calcium, Magnesium, Zinc, lead, etc occur as Carbonate.

d) Sulphate Ore

- Sodium, Calcium, Magnesium, lead, etc occur as sulphate.

e) Halide Ores

- Sodium, Calcium, Silver, occur as Halide.

f) **Phosphate Ores**

➤ **Calcium.**

▪ **The main mineral of Iron**

Mineral	Chemical Formula	Percentage of Iron	Ores
Magnetite	Fe_3O_4	12.4 %	Hematite
Hematite	Fe_2O_3	69.9 %	
Limonite	$\text{FeO}(\text{OH}) \cdot \text{H}_2\text{O}$	55 %	
Siderite	FeCO_3	48.2 %	
Goethite	$\text{FeO}(\text{OH})$	62.9 %	

• **Metallurgy**

➤ The process of extracting metal from their ores and refining them is known as metallurgy.

• **Principle of metallurgy**

➤ The extraction of metals from their ores depends on reactivity of the metal.

➤ High reactive metal: Electrolytic decomposition.

➤ Moderate reactive metal: Calcination.

➤ Least reactive metal: Heating.

▪ **The common term which are used in metallurgic operation**

○ **Charge**

➤ The mixture of material fed to be furnace for obtaining the metal is called charge.

○ **Gangue / Matrix**

➤ They are mined from the earth crust contain some unwanted substance [impurities] like as sand, rock, clay etc. These substance are called Gangue.

○ **Dressing of the Ores**

➤ The removal of impurities from the Ores is called Dressing of the Ore.

• **Calcination**

➤ The process of heating the Ores to a high temperature to convert the metal into its oxide either in the absence or insufficient amount of oxygen is called calcination.

➤ It is generally used to carbonate ores.

▪ **Roasting**

➤ The process of heating of finally , ground ores to a high temperature in the excess amount of oxygen (air) to convert ore to its oxide called roasting.

➤ It is used for sulphide Ores.

▪ **Flux**

➤ A flux is a substance that is mixed with the furnace charge [Calcinated / Roasted Ores] to remove the infusible impurities present in the Ores.

• **Slag**

➤ Since flux combine with the infusible impurities to convert them into fusible substance called slag.

▪ **Difference between Calcination or Roasting**

Calcination	Roasting
In this process, the Ores is heated in the absence of air.	In this process of Ores is heated in the presence of air.
It is used for Oxide and Carbonate Ores.	It is used for Sulphide Ore.
CO ₂ is produced along with metal oxide. $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$	SO ₂ is produced along with metal oxide. $2 \text{ZnS} + 3\text{O}_2 \rightarrow 2 \text{ZnO} + 2 \text{SO}_2$

Only for Understand

Infusible Substance	Flux	Fusible Slag
Not in molten State		
SiO ₂	+ CaO	→ CaSiO ₃
P ₂ O ₅	+ 3 CaO	→ Ca ₃ (PO ₄) ₂
MnO	+ SiO ₂	→ MnSiO ₃
		Molten State

➤ The Slag is lighter part which floats over the molten metal and is removed from there.

▪ **There are four metallurgical operations**

1. Dressing of Ores.
2. Convert dressed Ores into its Oxide.
3. Extract metal from metal Oxide.
4. Purification of metal.

1. **Dressing of Ores**

➤ The process of removing the impurities from the Ores is called Dressing of the Ores.

▪ **There are following process to remove the impurities from the Ore**

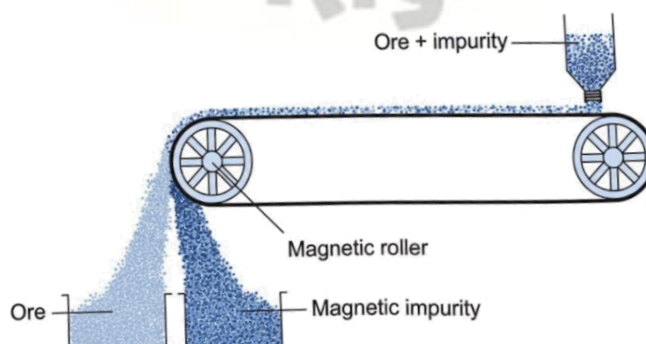
a) **Hand Picking**

➤ In this process we pick out the larger impurities from the Ore by hands like as sand, clay and rock or etc.

b) **Magnetic Separation process**

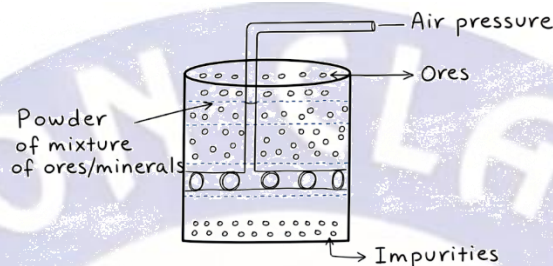
➤ This process is only applicable for those ores whose one of the component is magnetic in nature.

➤ First, Ores is converted into Ores powder Now, the Ores powder is falling on the moving belt which one part is magnetic wheel and another end is non – magnetic wheel due to which magnetic substance is attracted by magnetic wheel and collected in a container and non – magnetic substance are collected in other container away from the magnetic wheel. Thus, both the component are separated by this process.



c) **Froth – Floatation Process**

- This process is only applicable for those of the Ores in which impurities are heavier than Ores.
- Firstly, we convert Ores into ore powder and put into a container having water and few drops of pine oil. Now, adding pressure of air in the mixture is springing well. This process is continue after some time its lighter part comes out above the surface with miscells and heavy impurities settle down in the continuous for finishing the miscells then we adding some HCl and dry the Ores.



Note:

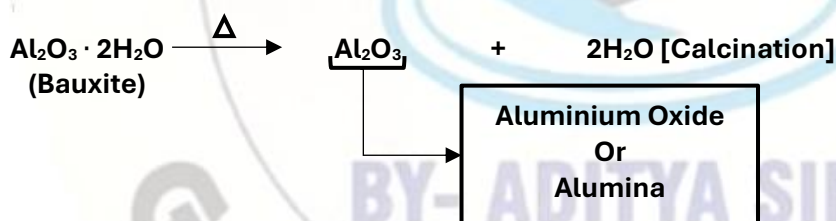
- Sulphide Ores of Copper, lead and Zinc are generally concentrated / dressing by this process.

2. **Conversion of Dressing Ores into metal Oxide**

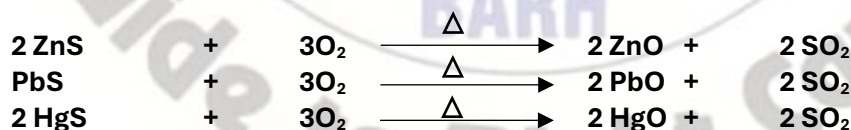
- The Dressing Ores are converted into metal oxide by either calcination or Roasting process.

For example:-

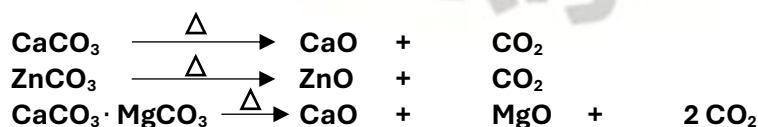
- ✓ Oxides Ores are calcined to remove moisture and other impurities.



- Zinc blende [ZnS] is roasted in air to convert into Zinc Oxide.



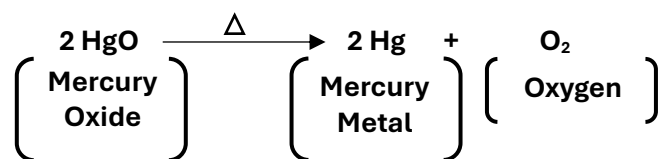
- Carbonate Ores are Calcinated to expel carbon dioxide



3. **Extraction of metal from metal Oxide**

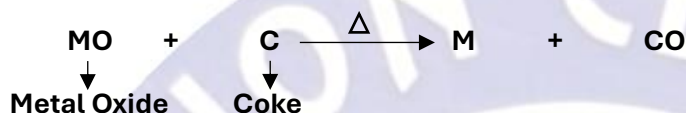
- It depends on their Reactivity.
- **Extraction of less reactive metal**
- The less reactive metal is extracted by the reduction of heat.

For example:-



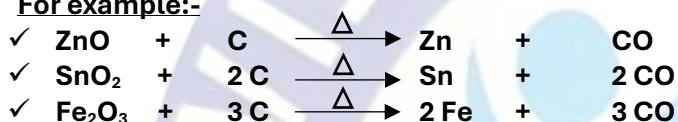
▪ **Extraction of Moderately reactive metal**

- The metal present in middle of the reactivity series are extracted by the reduction of their oxide with carbon, Aluminium, sodium or calcium.



- Reduction of metal Oxide with carbon is known as Smelting.

For example:-



Note:

- Some metal Oxide cannot be reduced by carbon because it combines with carbon to form carbide.

▪ **Extraction of metal Oxide with Aluminium**

For example:-



• **Thermite Reaction**

- The reduction of metal oxide to form metal by using Aluminium powder as reducing agent is called thermite reaction [Thermite process].
- A mixture of Iron (III) oxide and Aluminium powder is ignited with a burning magnesium ribbon. Aluminium reduces iron oxide to produce iron metal with the evolution of lot of heat. Due to this heat, iron metal is produced in molted salt.



- The molten iron is then poured between the broken iron piece to weld them [to join them]. The process is called alumino thermy or thermite welding. Thus, Thermite welding making uses of the reducing property of Aluminium.

For example:-

- ✓ For joining the rails.



▪ **Extraction of highly reactive metal**

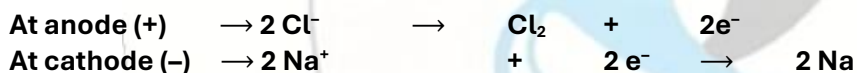
- The highly reactive metal are extracted by electrolytic reduction of their molten oxide or chloride.
- In this process cathode supplied electrons to metal ions for their reduction to the metal.

Note:

- Electrolytic decomposition is used in molten Ores.

For example:-

- ✓ Sodium is obtained from molten sodium chloride.



Finally,



- ✓ **Magnesium metal is obtained by molten magnesium chloride.**



4. Purification of metal

- The metal obtained from the Ore is not pure. It contains various substance are impurities. The process of removing these impurities is called purification of metal.
- **There are various method of purification of metal**

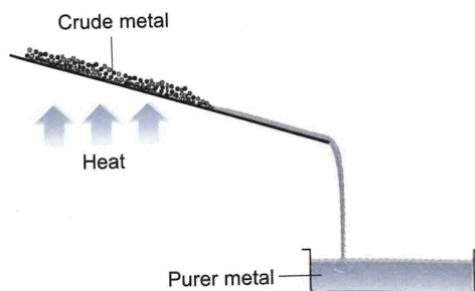
a) **Liquation**

- This process is used to separate metals of low melting point from the metal of high melting point.

For example:-

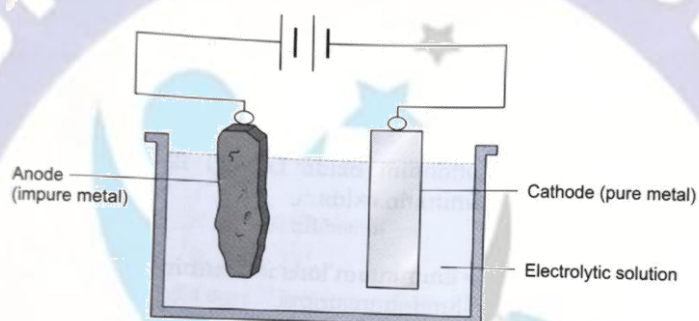
- ✓ Tin & Lead.

- In this process a sloping hearth is use. The hearth is kept at a temperature little above the melting point of the metal. The impure metal is placed at the top of the hearth. The metal melts and flow down the hearth, The infusible substance are left behind.



b) Electrolytic Refining

- This method is widely used for purification of metal such as Aluminium, Copper, Tin, Lead, Gold, Zinc & Chromium.
- The impure metal is made of Anode while, a strips of pure metal acts as the cathode. A solution of salt of the metal is used as electrolyte.

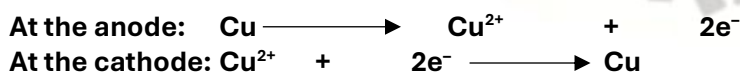


Note:

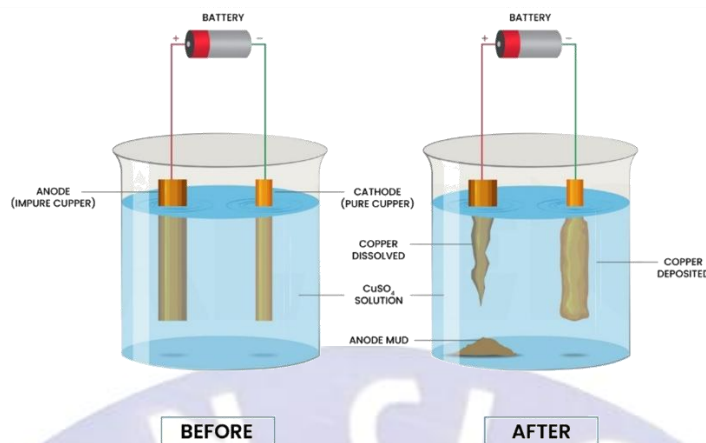
- On passing electric current through the solution pure metal gets deposited on the cathode. The more reactive impurities present in the metal to remain their the less reactive impurities fall to the bottom of the electrolytic cell at Anode which is known as anode mud.

For example:-

- ✓ **Electrolytic Refining of Copper**
- Electrolytic cell containing an electrolyte of Copper sulphate solution and dilute sulphuric acid with anode of blistered copper and cathode of thin sheets of pure copper is used on electrolysis copper passes into solution from the anode and the equivalent amount of pure copper is deposited from the solution at the cathode.



Then, the obtained copper is 99.99 % pure.



Note:

- The impurities like bismuth, iron, lead, and nickel being more basic than copper pass into solution and are not discharge at the cathode while gold, silver & platinum are collected as anode mud.

- **Electrolyte**

- A liquid / gel which contains ions called electrolyte.

Example of metal Salt:-

- ✓ CuSO_4 , $\text{Al}_2(\text{SO}_4)_3$, FeSO_4 etc.

- **Some metals their Ores and extraction of metal from their Ores.**

Iron

- ➔ Latin word → Ferrum
- ➔ Symbol → Fe
- ➔ Atomic no. → 26

Important Ores

- ✓ Haematite (Fe_2O_3)
- ✓ Magnetite (Fe_3O_4)
- ✓ Limonite ($2\text{FeO}_3 \cdot 3\text{H}_2\text{O}$)
- ✓ Siderite (FeCO_3)
- ✓ Iron pyrites (FeS_2)

- Iron is mostly extracted by Hematite Ores.

Zinc

- ➔ Symbol → Zn
- ➔ Atomic no. → 30

Important Ores

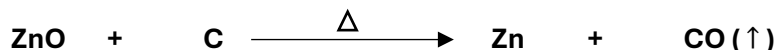
- ✓ Zinc blende (ZnS)
- ✓ Calamine (ZnCO_3)

Case 1: Extraction of Zinc from Zinc blende

- When zinc blende is roasted in the sufficient amount of air, then zinc oxide is formed.



- When zinc oxide is heated with coke then zinc is obtained.



Case 2: Extraction of Zinc from calamine.

- When calamine is calcined then we obtain zinc oxide & carbon dioxide.



- When zinc oxide is heated with coke then zinc is obtained.



Aluminium

- Symbol → Al
- Atomic no. → 13

Important Ores

- ✓ Bauxite [$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$]
- ✓ Cryolite [Na_3AlFe]
- ✓ Corundum [Al_2O_3]
- ✓ Feldspar or Felspar [KAlSi_3O_8]

- **Aluminium is mainly extracted by Bauxite Ore by Bayer process**

- In this method, Powdered bauxite ore is treated with hot sodium hydroxide solution. The sodium hydroxide reacts with Aluminium oxide present in bauxite ore, to form water soluble sodium aluminate.
- The impurities are removed by filtration.



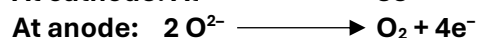
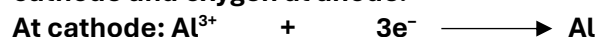
- After dilution and removal of precipitate impurities the liquor is stirred with crystal of Aluminium hydroxide when most of the Aluminium hydroxide $\text{Al}(\text{OH})_3$ is precipitated.



- The precipitate of $\text{Al}(\text{OH})_3$ is filtered, washed dried and heated strongly to get pure alumina.



- Now, we pass electricity in the molten state of alumina. In this process Aluminium is decomposed at cathode and oxygen at anode.

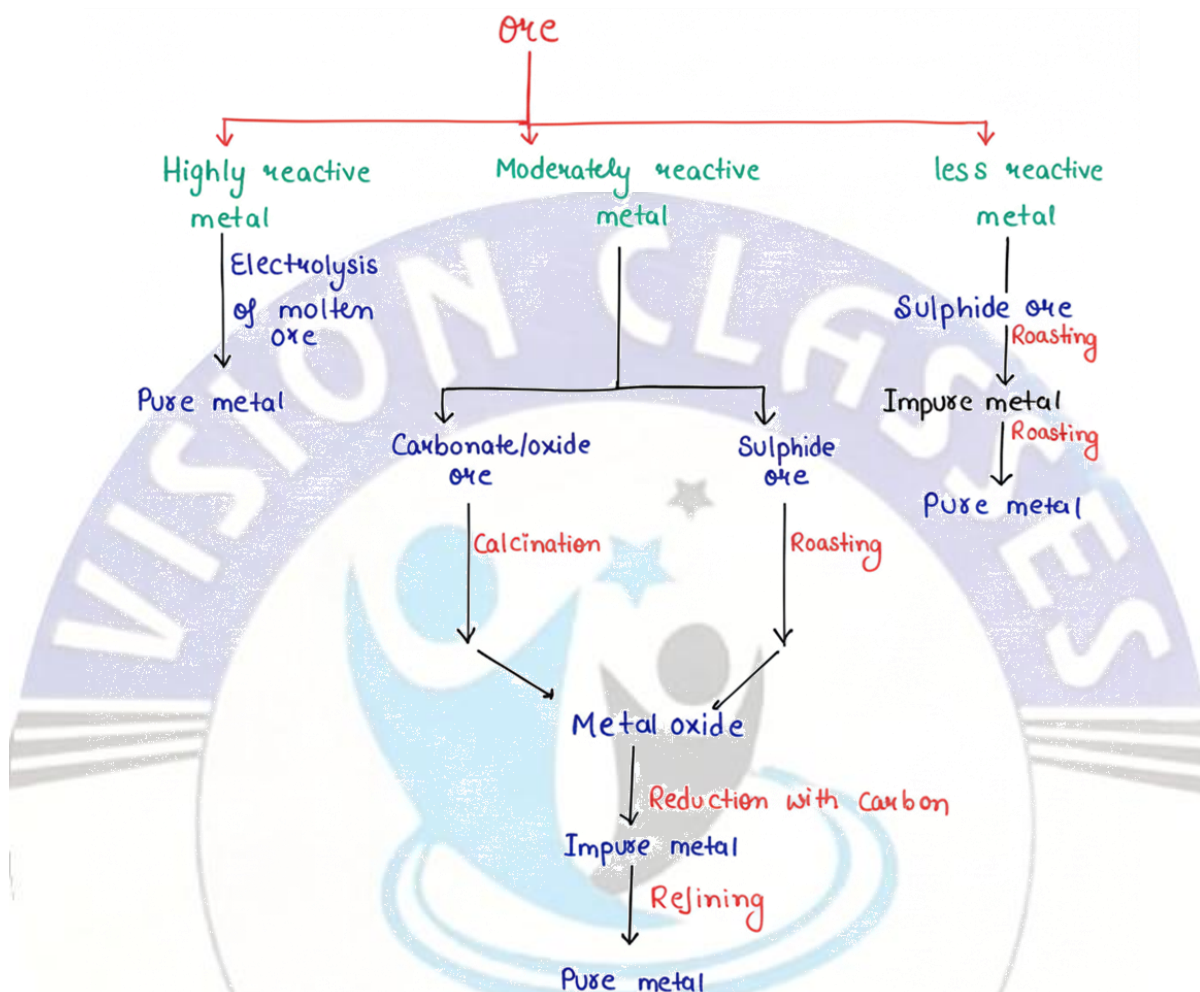


Mercury

- Symbol → Hg
- Atomic no. → 80

Important Ores

- ✓ Cinnabar [HgS]



• Corrosion of metal

- When metal is left in the air for a long time then it reacts with moisture, air or chemicals etc. Then metal are eating up on their surface is called corrosion.

For example:-

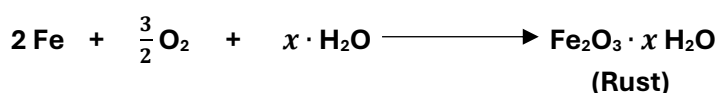
- ✓ Rusting of iron, corrosion of copper, Aluminium etc.

• Rusting of Iron

- When iron article are left in the air for a long time then it reacts with air, moisture, chemical etc. due to which a brown layer is deposited on the surface of iron known as Rusting of Iron.
- Corrosion of Iron is continuous process which ultimately eats up the whole iron objects.

Note:

- The corrosion of other metal is called Corrode but Corrosion of Iron called Rusting.



▪ **Condition of Rusting of Iron**

- ✓ Presence of air.
- ✓ Presence of water.

Note:

- Damp air alone supplies both the thing air or water.

▪ **The process of prevention of rusting of Iron**

- a) Applying paint on the surface of iron particle.
- b) Applying grease / Barnish or oil on the surface of iron particle.
- c) Rusting of iron can be prevented by galvanisation.

• **Galvanisation**

- The process of depositing a thin layer of zinc metal on iron object is called galvanisation.
- d) Rusting of iron can be prevented by tin plateting / cromium plating.
- e) Rusting of iron can be prevented by converting it into its alloy.

• **Alloy**

- An alloy is a homogeneous mixture of two or more metals or a metal and non – metal.

For example:-

▪ **Amalgam**

- An amalgam is an alloy of mercury and one or more metals.
- Amalgam may be solid or liquid.
- Amalgam of sodium & Aluminium are good reducing agents.
- Amalgam of Silver, tin, cadmium and copper have been as dental filling.

▪ **Alloy of steel**

- Steel is an alloy of Iron and carbon in the composition of carbon contain 0.15 – 17%.

▪ **Stainless steel**

- Stainless steel is an alloy of iron, carbon, chromium and Nickel.
- In stainless steel, 11 to 12% of chromium, Nickel = 8%, Carbon = 0.08%, Iron = 68% to 78%.
- Stainless steel does not rust or stain.
- It is used in the purpose of Industrial & Domestic fields.

▪ **Some alloy & their composition & user**

<u>Alloy</u>	<u>Composition</u>	<u>Uses</u>
Solder	Sn = 50%, Pb = 50%	For joining metal soldering electronic components etc.
Duralumin	Al = 95.5%, Cu = 3%, Mn = 1%, Mg = 0.5%	In bodies of aircrafts, kitchen utensil, automobile parts.
Bell metal	Cu = 78%, Sn = 22%	Bells, gongs etc.
Manganin	Al = 84%, Mg = 12%, Ni = 4%	Standard resistors.
Pewter	Sn = 90%, Pb = 5 – 10%, Cu = 2%	Cups & mugs etc.
Bronze	Cu = 90%, Sn = 10%	To build Statue, medals.
Brass	Cu = 70%, Zn = 30%	Door handles, Statue.

Alloy of Steel	Composition	Properties	Uses
	Iron = 75% Cr = 17% Nickel = 4% Copper = 4%	Extremely hard, resistant to Corrosion.	Grinding machines, Safes, railing, gates, etc.

▪ **Properties of Alloy**

- a) They are harder than their constituent but less ductile and malleable.
- b) They are resistance to Corrosion.
- c) The melting point of alloy may be higher and lesser than any of its constituents.
- d) The properties of an alloy are much more improved and pronounced than those of its constituents.

For example:-

- ✓ Aluminium is not very strong but its alloy duralumin is light and very strong.

- e) The copper of an alloy is different from the metals from which it is formed.

• **Anodizing**

- It is a process of producing an oxide coating on the surface of a metal by electrolytic process.
- Anodizing is a process of forming a thick layer of Aluminium oxide on an Aluminium object by making it anode during the electrolysis of dilute sulphuric acid.
- Anodizing is used to prevent the corrosion of Aluminium.

For example:-

- ✓ An Aluminium article can be Anodized by making it the anode during electrolysis. Oxygen gas evolved at the anode reacts with the metal forming a thick layer of oxide of the metal.



BY- ADITYA SIR
BARH

Guide to Right Career